

WhisperWard OSINT: Case Study

An explainable digital investigations platform for Trust & Safety and child-protection investigations.

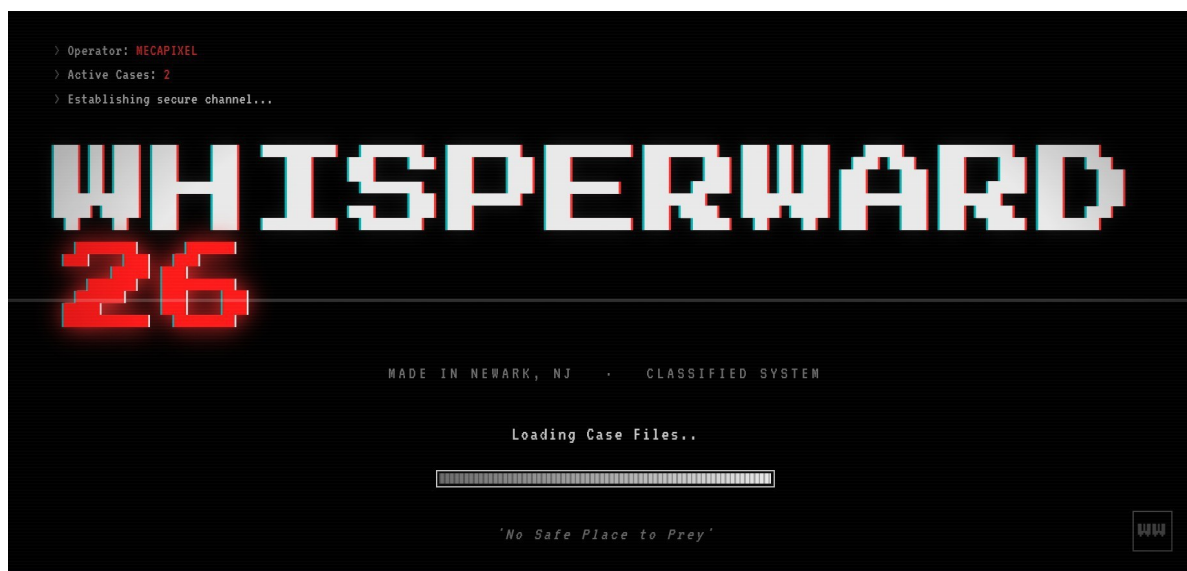
Author: Meca Dismukes **Repository:** github.com/Mecapixel/whisperward-osint

Live demo: whisperward-osint.onrender.com **License:** AGPL-3.0

Overview

WhisperWard is a deployed, open-source intelligence (OSINT) platform designed to support investigations into online child exploitation on gaming and social platforms. I designed and built the entire system end-to-end. It guides investigators from initial subject discovery through to court-defensible evidence packaging by combining automated public data collection, cross-platform identity correlation, analyst-gated identity resolution, a fully transparent weighted risk-scoring engine, and forensic evidence handling, all backed by mandatory human review at its core.

WhisperWard runs a live public demonstration environment, carries 659 automated tests in continuous integration, and is published under AGPL-3.0 so its safeguards remain open and auditable. Since first release it has been evolved through a four-phase platform refactor: the investigation machinery now lives in a reusable core, with child safety as its first specialization module, and the platform speaks industry standards through STIX 2.1 export and an honest MITRE ATT&CK mapping. It has since gained a persistence layer that tracks account reconstitution, and a documented methodology that tests the risk engine against its own claims.



WhisperWard boot screen — operator shown as unauthenticated until sign-in, with the live active-case count

The Problem

Online child exploitation on platforms like Roblox and Discord continues to grow. These environments are primary spaces for children to socialize, and predators heavily target them for grooming.

Investigators in law enforcement, platform Trust & Safety, and nonprofits face significant challenges: grooming signals are fragmented across platforms and time, and existing tools often lack the ability to correlate data and produce defensible, explainable assessments.

Four core constraints shaped WhisperWard's architecture:

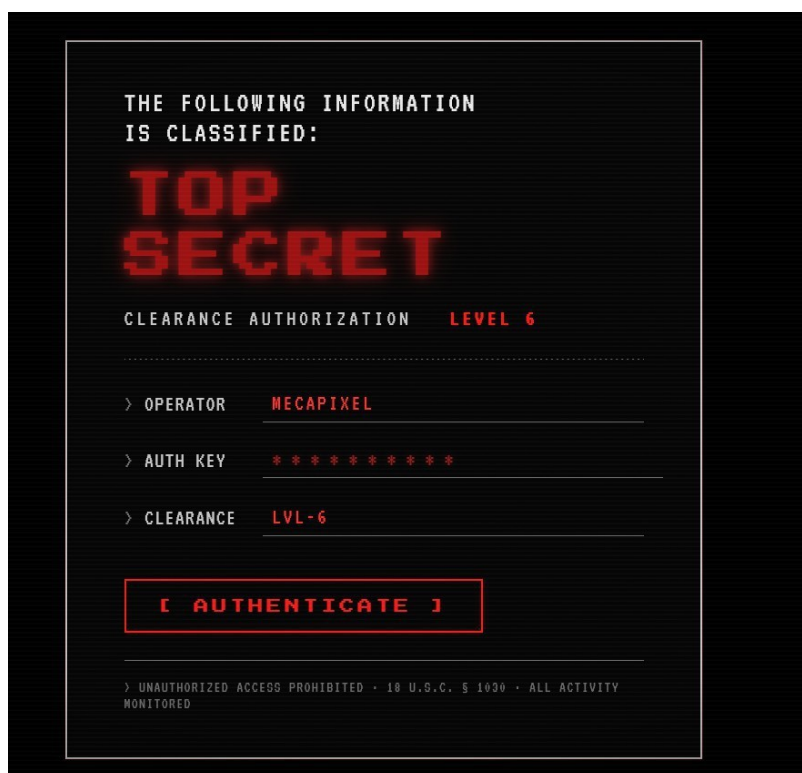
- **Fragmented, cumulative signals.** Grooming is rarely evident in a single message. It emerges as patterns such as age probing, boundary testing, secrecy requests, and platform migration, unfolding across time and accounts. The tooling must reason about the pattern, not the message.
- **Explainable, defensible assessments.** Any system that may inform referrals or legal processes must let reviewers trace every risk point to its source. A black-box score is unacceptable in this domain.
- **Human authority over identity claims.** Saying two accounts are the same person is a consequential claim. The machine may propose it with evidence; only a named analyst may resolve it, and that decision must be permanently attributable.
- **Enforcement measured at the operator, not the account.** Removing an account ends that account, not the behavior behind it. An operator who returns within days has defeated the enforcement while the record shows a successful takedown. Measuring disruption requires modeling which account replaced which, and how quickly.

What WhisperWard Does

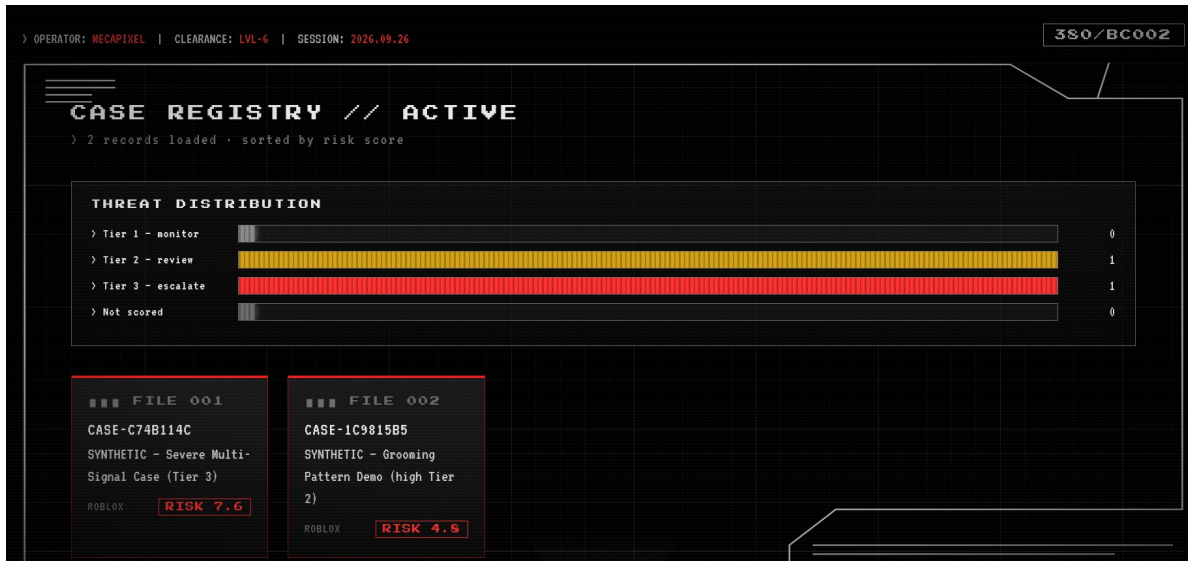
The platform executes a complete investigative pipeline:

1. **Collection.** Gathers public profile data from platforms like Roblox and correlates usernames across sites.
2. **Correlation.** Links identities using fuzzy matching, writing-style analysis, and perceptual avatar hashing, exposed through a first-class CLI correlate command.
3. **Identity resolution.** Proposes candidate entities from correlation output with full per-membership justification; promotion to a resolved identity requires a named analyst and is sealed into the chain of custody. A justified identity graph and a strictly reconstructive investigation timeline sit alongside it.
4. **Persistence tracking.** Orders removed and successor accounts into directed reconstitution chains, with the return interval on every link, so enforcement outcomes can be measured rather than assumed.
5. **Behavioral analysis.** Detects grooming patterns in chat content, accounting for multi-step sequences.

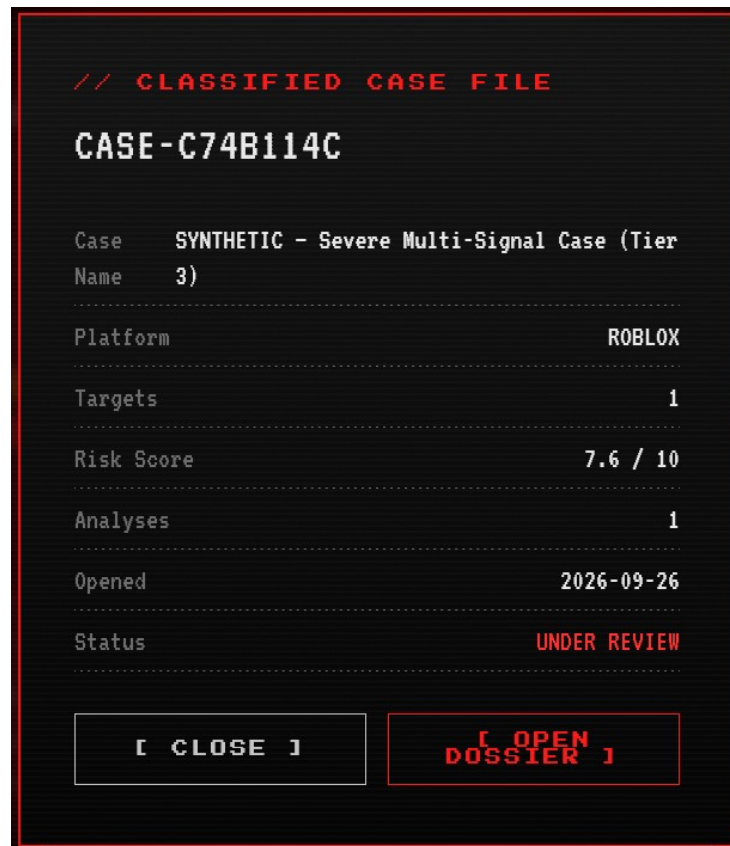
6. **Risk scoring.** Generates a weighted, fully explainable risk score with tier recommendations, reasoning over graph-corroborated platforms when identity-graph inputs are present.
7. **Evidence & reporting.** Packages artifacts with SHA-256 hashing and chain-of-custody manifests, supports redaction and referral export, and exports cases as byte-stable STIX 2.1 bundles with an ATT&CK technique mapping.
8. **Human review.** Every escalation requires human approval. The system provides evidence and reasoning. It never makes autonomous decisions.



Operator sign-in — demonstrates the operator workflow; the public demo holds synthetic data only and is not access-controlled



Case registry — active cases sorted by risk, with the D3 threat-tier distribution



Classified case file — quick-view popup

The Explainable Risk Engine (Core Differentiator)

The risk engine is the heart of WhisperWard. Instead of opaque scores, it evaluates subjects across five weighted components, each with traceable contributions and plain-language explanations:

Component	Weight	What it measures
Grooming language	40%	Multi-category pattern detection in chat content, language only
Cross-platform correlation	25%	Same identity detected across multiple platforms
Anonymization	15%	Use of Tor / VPN infrastructure
Behavioral velocity	10%	Account age and rapid friend acquisition
Historical signals	10%	Prior flags in the investigation database

A capped synergy term rewards independent signals that occur together, and it is itemized with its reasons like every other part of the score. Scores map to three tiers: Tier 1 (Monitor), Tier 2 (Human Review Required), and Tier 3 (Escalate). Escalation to Tier 3 requires more than a score of 7.0: it also requires cross-platform presence or prior history. A score that reaches 7.0 on single-platform evidence alone is held at human review with the reason recorded. This requirement is enforced as an explicit rule, not left to the arithmetic of the weights, because analysis showed the weights alone could reach exactly 7.0 on a single platform.

When identity-graph inputs are present, the cross-platform component reasons over graph-corroborated platforms, and a contradiction on any supporting edge caps confidence without ever altering the score itself — uncertainty is surfaced, never hidden inside the number.

The result: every score is fully accountable, and the parts shown always add up to the number shown. A reviewer sees not “7.6,” but “7.6 because grooming language contributed 3.39 points, cross-platform presence 2.50, behavioral velocity 0.70, and corroboration synergy 1.00,” each with its own plain-language explanation.



From Tool to Investigation Platform

After first release, WhisperWard was deliberately re-architected across four platform phases, growing the test suite from 424 to 659 along the way, green at every milestone.

- **Core extraction.** The investigation machinery — contracts, registry, plugin base, correlation, scoring, evidence, redaction, retention, and reporting — was separated into a reusable core. Child safety became the first specialization module, registering its collectors, classifiers, and behavioral indicators through a registration seam. The core never imports a specialization, so the same engine can support other Trust & Safety domains.
- **Evidence and explainability engine.** Every score component and result carries a confidence level with enumerated reasons that never alter the score. Findings are promoted to queryable explanation objects linked to their evidence, analysts annotate through an append-only notes layer chained into the custody log, and the full case lifecycle leaves a tamper-evident audit trail.
- **Entity resolution and identity graph.** The resolver proposes candidate identities from correlation output; a contradiction on any supporting edge excludes an account with the reason recorded. Promotion requires a named analyst and lands in the chain of custody — unattributed resolution refuses to run. The identity graph carries a complete justification on every edge, answers path queries with the evidence behind them, and serializes deterministically for evidence packaging. The investigation timeline is strictly reconstructive: every event names its source, and no event is ever inferred.
- **Standards-aware intelligence.** Cases export as byte-stable STIX 2.1 bundles — user-account observables, correlation relationships with confidence and rationale, analyst-resolved entities, and a suspicious-activity grouping — that deliberately never emit adversary-assertion object types. The MITRE ATT&CK mapping is honest by construction: it maps only fired technical signals, grades each mapping direct or analogue, and states which findings fall outside the framework's scope and where they are documented instead. A written threat model covers the platform's own assets, adversaries, failure modes, and residual risks.

Methodology: Testing the Engine Against Its Own Claims

An engineering review asked the questions any technical interviewer would: where the weights came from, how robust the scoring is, and whether a subject could deliberately evade it. The answers are documented in docs/METHODOLOGY.md, and every figure in that document is checked by the test suite.

- **Where the parameters come from.** Tier thresholds were calibrated by a threshold sweep against a seeded synthetic dataset. Classifier category

weights are ordered by the grooming literature. The five component weights are an expert-judgment prior, ordered by how directly each component observes the behavior of concern, and stated as such rather than presented as fitted. Fitting them would require labeled case outcomes that an independent researcher cannot lawfully hold.

- **How sensitive the score is.** Across a 1,536-profile grid covering every combination of inputs, moving any single weight by ten percent changes the tier of at most 2.7 percent of profiles, and a quarter-sized change never moves a score by more than one point. Tier output does not depend on the exact weight values, which is the real defense of hand-set weights.
- **How it fails.** Burying grooming language in benign chat has no effect. Dropping anonymization, using an established account, or spacing tactics out each moves a subject from escalation to human review, never to monitor-only. Rewording the same intent defeats the lexical classifier, and a subject combining every technique is not detected. The engine is a triage aid for patterns it models, and it is presented as one.
- **What it measures.** On the synthetic benchmark, the engine flags no safe profile, catches 45 of 50 threat profiles, and scores an F1 of 0.87, all enforced by tests. Because those profiles were written alongside the classifier, the figures establish internal consistency, not real-world detection, and the documentation says so.

The analysis also found two places where the engine did not do what its documentation said. The Tier 3 guarantee described above held only by coincidence of arithmetic, and account age was counted twice, once in velocity and again inside the grooming score. Both were corrected in the engine. The severe demonstration case moved from 8.2 to 7.6 as a result: its grooming score had been inflated by its friend count, which says nothing about its language. The lower number is the more defensible one, and the weights were deliberately not re-tuned to restore the earlier results.

Measuring Enforcement: The Persistence Graph

Correlation answers whether two accounts appear to be operated by the same person. It does not answer whether one replaced the other after a removal — and that directed, temporal relationship is what shows an enforcement action failed. The identity graph could not express it: it stores edges under a sorted pair, so direction is discarded by design.

The persistence layer sits on top of that graph rather than replacing it. It holds a directed graph of succession and cross-platform handoff edges, reusing the identity graph's justification model so an edge without a rationale cannot exist at all.

- **Direction is derived, not asserted.** Order comes from removal and creation timestamps, never from the correlation, which carries no ordering. Cycles are refused, since a succession chain is temporally ordered and a cycle means the ordering is wrong.
- **Inference is held to a floor.** Succession is inferred from correlation output only above a configurable strength threshold and inside a reconstitution

window, skipping any pair carrying a contradiction. Every derived link is flagged as inferred, so analyst-observed successions stay separable from suggested ones.

- **Return intervals are the measurement.** Each link records how long the operator took to come back, with a median per chain. A return measured in hours is not a coincidence, and it is the number that makes a pattern legible to a platform.
- **Two export views.** A full-fidelity referral view retains identifiers for the lawful recipient. A platform-facing view replaces every account with an opaque chain and position label while preserving chain length, intervals, and platform span — carrying an explicit note that chains are correlation groupings, not identifications of natural persons.

Reconstitution chains with return intervals are an original analytic contribution, built to answer a question the existing tooling does not ask. A companion project, Recidivism Analytics (github.com/Mecapixel/recidivism-analytics), studies the same return-after-ban behavior at population scale on a synthetic dataset that retains ground truth, so its detection claims can be audited for precision and recall.

A Defect the Test Suite Could Not See

Wiring the persistence layer to the command line produced a result worth recording. With the module complete and its full test suite green, the first run against real case data inferred nothing — at any threshold, with no error explaining why.

The cause was a sentinel mismatch. The correlation engine recorded a clean comparison by writing the sentence “no contradictions detected” into its contradiction field, while every downstream consumer tests that field for emptiness. A clean pair therefore read as contradicted, and every pair was discarded. The unit tests passed throughout, because they construct their own justifications using the empty-string convention rather than the engine's real output — the defect lived precisely in the gap between the two.

Tracing it through version history showed the sentinel was introduced with the extracted correlation engine and became live one day later, when the identity graph first consumed the field — roughly eight weeks before discovery. A full audit of consumers during the root-cause fix showed the reach was wider than the first assessment: the identity graph reported zero contradiction-free lead edges and no corroborated platforms; graph-aware confidence was capped at medium on every case, with an inaccurate reason recorded; the entity resolver never saw a clean lead from real engine output; STIX relationships carried a false “contradiction observed” clause; and the command line marked every edge as contradicted.

The failure direction matters. In every one of those places the platform understated its own evidentiary support rather than overstating it, which is the correct direction to fail in work that may inform a referral.

The fix landed in two deliberate steps. The persistence commit patched its own consumer with tests covering the sentinel directly, so the feature could ship. The

root cause was then corrected in an isolated commit (fa8a4de), because it moves confidence labels across every existing case and that movement should be visible on its own. The engine now emits an empty string for a clean pair; a single shared module defines what “clean” means; and every consumer normalizes the field at ingest, so correlation records sealed before the fix are read correctly without rewriting evidence. Eight regression tests push real engine output — not hand-built fixtures — through the graph, the resolver, and the persistence layer, and four of them fail against the pre-fix code.

The broader lesson is recorded here on purpose: a green suite demonstrates that components behave as their tests assume, not that they agree with each other. Integration against real data remains the only check on the assumptions the tests themselves encode. The same lesson later surfaced in the demonstration itself, whose pre-computed scores had silently drifted from the engine's text; they are now generated from the engine, and a test fails if they ever differ.

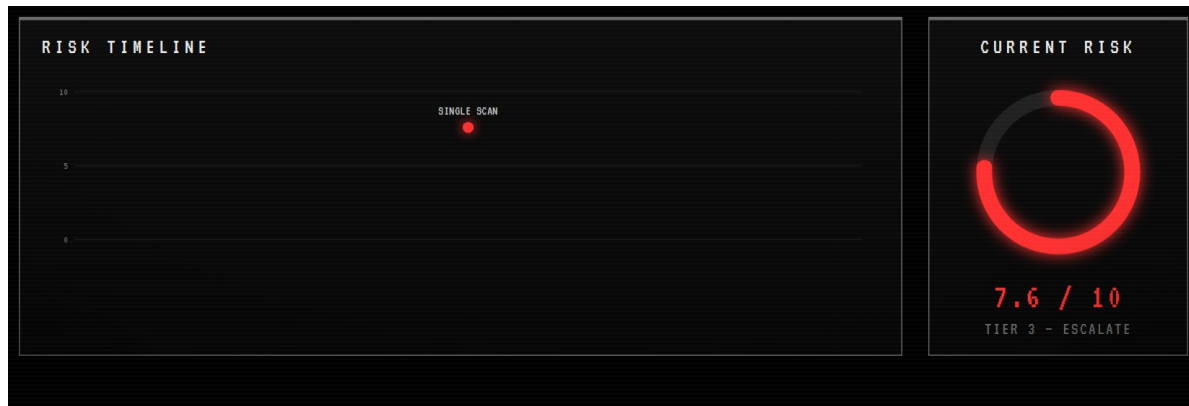
Demonstration

The public demo carries two synthetic cases, built from clearly labeled fabricated material, which the project's governance framework permits — no real minor's data and no third-party accounts are processed. They exercise the engine across its range and show that it discriminates severity based on real signals rather than flagging everything alike.

- **Tier 2, Human Review Required (score 4.8).** A synthetic subject exhibiting a grooming-language pattern (age probing, a compliment, platform-migration pressure, and a secrecy request) across two platforms on a young account. The engine flags it for review without over-escalating, which is the realistic, common case.
- **Tier 3, Escalate (score 7.6).** A synthetic subject exhibiting a strong multi-category grooming pattern across four platforms on a very new, high-velocity account. Every axis fires, the corroboration synergy applies, the four-platform footprint satisfies the corroboration rule, and the engine escalates to an evidence package with mandatory human sign-off.



Synthetic case dossier (Tier 3, score 7.6) with explainable scoring



Risk timeline and current-risk gauge (Tier 3, score 7.6)

The contrast between the two demonstrates the design goal: the score is not a gut feeling, it is a defensible sum of documented parts, and the same engine correctly separates a review-level case from an escalation-level one.

Governance, Ethics, and Safety

Safeguards are foundational, not afterthoughts:

- **Human-in-the-loop by design.** No autonomous decisions. Every escalation requires human sign-off before any action, and resolving two accounts into one identity requires a named analyst whose decision is sealed into the custody chain.
- **Approval-gated CSAM handling.** Perceptual-hash detection is disabled by default and gated behind explicit approval and credentials available only to vetted organizations. It never stores image content, processes hashes in memory only, and flags every match for mandatory human review.

- **Synthetic data only.** All demonstration and testing uses clearly labeled fabricated data and investigator-created test accounts. No real minor's data and no third-party accounts are processed.
- **Operator attribution, not access control.** The dashboard's sign-in records a named operator, and the authoritative operator record lives in the tamper-evident chain-of-custody log. In the public demo, which holds only synthetic data, the sign-in illustrates the workflow and is not a security boundary. A deployment holding real case data would add credential authentication and access control on every route, including the API.
- **No adversary assertions in exports.** The STIX 2.1 export shares observables, relationships, and analyst-resolved entities with their confidence and rationale — it never emits object types that would assert someone is a threat actor. That determination belongs to humans and due process.
- **Persistence chains are groupings, not identifications.** Reconstitution output carries an explicit statement that a chain describes accounts believed to be operated in coordination, and that inferred links are suggestions for an analyst to confirm. The platform-facing view withholds identifiers entirely.
- **Bias standard defined before real data.** Governance defines a release gate for any deployment that processes real accounts: a demographic proxy group with a false positive rate more than five points above baseline blocks the release. The synthetic-only build does not yet run these audits, and the documentation states that plainly.
- **Open and auditable.** Published under AGPL-3.0 specifically so the safeguards cannot be quietly stripped out.

Engineering

- **Stack.** Python, FastAPI, SQLite, D3.js, NetworkX, and stix2, with a modular, plugin-based collection architecture.
- **Quality.** 659 automated tests running in GitHub Actions on every push. The scoring engine, behavioral classifier, correlation, entity resolution, identity graph, persistence graph, and STIX export modules are independently tested; the contradiction seam is covered with real engine output; benchmark targets, methodology bounds, and demo-to-engine parity are enforced; and the suite runs clean from a fresh clone.
- **Architecture.** A reusable investigation core with child safety as its first specialization, connected through stable contracts and a registration seam; an explainable scoring engine decoupled from data collection so scores are reproducible and traceable; and a web layer that reads the same persisted data the command-line tools produce, so the dashboard and the reports always tell the same story.
- **Interface.** A full command-line tool for investigators — including first-class commands for correlation, entity proposal and promotion, the identity graph, the timeline, account state and reconstitution chains, STIX export, and ATT&CK mapping — plus a web dashboard for review and visualization with risk-timeline, tier gauge, and explainable breakdown views.

What's Next

The platform build is complete: all four platform phases have shipped, along with the correlation CLI wiring, the persistence layer and its command-line surface, the methodology and benchmark enforcement, and a tokenless Discord public-OSINT module on the plugin contract. Remaining directions are additive:

- **Persistence in the dashboard.** Reconstitution chains are available from the command line; surfacing them in the web interface, alongside automatic capture of removal and creation timestamps during collection, is the next increment.
- **Semantic behavioral analysis.** The methodology shows the lexical classifier is defeated by paraphrase. Evaluating small, fine-tunable language models as an additional signal, kept as an explainable component rather than an opaque score, is the direct response.
- **Production access control.** Credential authentication and route-level authorization, including the API, before any deployment holds real case data.
- **Deeper platform coverage.** Extending collection and correlation to additional platforms central to how minors socialize online, each built on the existing plugin contract with public data only.
- **PhotoDNA / NCMEC activation.** The CSAM hash-detection architecture is complete and tested; activation waits on external authorization, exactly as the governance framework requires.
- **Collaboration.** Engaging with child-safety organizations and platform Trust & Safety teams that build and use tooling in this space, including partners able to supply the labeled outcomes needed to fit the weights empirically.

Summary

WhisperWard demonstrates the ability to take a hard, sensitive problem and build a complete, responsible solution — and then to evolve it like a platform: robust data collection, transparent risk assessment, analyst-gated identity resolution, measurable enforcement outcomes, standards-aware intelligence export, forensic evidence handling, and strong ethical guardrails. It is public, deployed, thoroughly tested, and honest about its own limits, which is the standard this work demands.

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